PROJECT REPORT | CLOUD INFRASTRUCTURE

Azure Disaster Recovery Project

Cross-region VM replication and failover with Azure Site Recovery

Primary region	West US 3
DR region	East US 2
Platform	Microsoft Azure – Azure Site Recovery
Report date	25 October 2026
Prepared by	_____

Table of Contents

Table of Contents 2

1. Executive Summary 3

2. Project Objectives and Scope 3

3. Solution Architecture 4

4. Azure Services and Components 4

5. Implementation Steps 6

6. Timeline and Results 14

7. Failover and Failback Runbook 15

8. Observations and Recommendations 15

9. Cost and Security Considerations 16

10. Challenges and Lessons Learned 16

11. Conclusion 16

Appendix A – Glossary 18

1. Executive Summary

This project designs, implements and validates a **disaster recovery (DR) solution on Microsoft Azure** using **Azure Site Recovery (ASR)**. A Linux virtual machine (**VM-1**) hosting an nginx web application runs in the primary region **West US 3**. It is continuously replicated to the secondary region **East US 2** through a Recovery Services vault, so the workload can be brought back online if the primary region becomes unavailable.

After the initial synchronisation, VM-1 reached the **Protected** state with an observed **Recovery Point Objective (RPO) of 31 seconds**. A **test failover** was then executed: ASR created the test VM **VM-1-test** in East US 2 in about two minutes, and nginx was verified on it (`systemctl is-active nginx` returned *active* and `curl -I http://localhost` returned *HTTP/1.1 200 OK*). The test environment was then removed by a **test failover cleanup**, and the replicated item returned to a Healthy, Protected state with no impact on production.

The result is a validated, repeatable DR capability that can be extended with recovery plans, automated DNS cut-over and monitoring.

Item	Result
Workload protected	VM-1 (Linux, nginx web server), Standard_D2s_v3
Primary / DR region	West US 3 / East US 2
Replication state	Protected, replication health Healthy
Observed RPO	31 seconds (as on 10/25/2026, 3:13:58 AM)
Test failover	Successful; test VM VM-1-test created and started in about 2 min 15 s
Application check on DR VM	nginx active; HTTP 200 OK
Test failover cleanup	Successful; test resources removed (about 1 min 11 s)

2. Project Objectives and Scope

2.1 Objectives

- Protect a production-style IaaS workload against a regional outage in Azure.
- Replicate the VM and its disk to a second Azure region with a low RPO.
- Prove recoverability by performing a non-disruptive test failover and validating the application.
- Document the procedure so that failover, failback and future DR drills are repeatable.

2.2 In scope

- Primary environment: resource group **Recovery** with VM-1, virtual network, public IP, NSG, NIC and OS disk.
- DR environment: resource group **DR_Recovery** with the Recovery Services vault, target virtual network (DR-V-NET), cache storage and replica disk.
- Enabling replication, monitoring replication health, test failover, validation and cleanup.

2.3 Out of scope (future work)

- Unplanned production failover and failback (re-protect) execution.
- Multi-VM recovery plans, automation runbooks, and DNS / Traffic Manager cut-over.
- Azure Backup policies for point-in-time data recovery.

3. Solution Architecture

The solution follows the standard ASR region-to-region pattern for Azure VMs. The ASR Mobility agent on VM-1 (version 9.67.7849.1, status Healthy) sends changes to a **cache storage account** in the source region; ASR processes them and applies them to a **replica managed disk** in the target region. The Recovery Services vault orchestrates replication, recovery points and failover jobs.

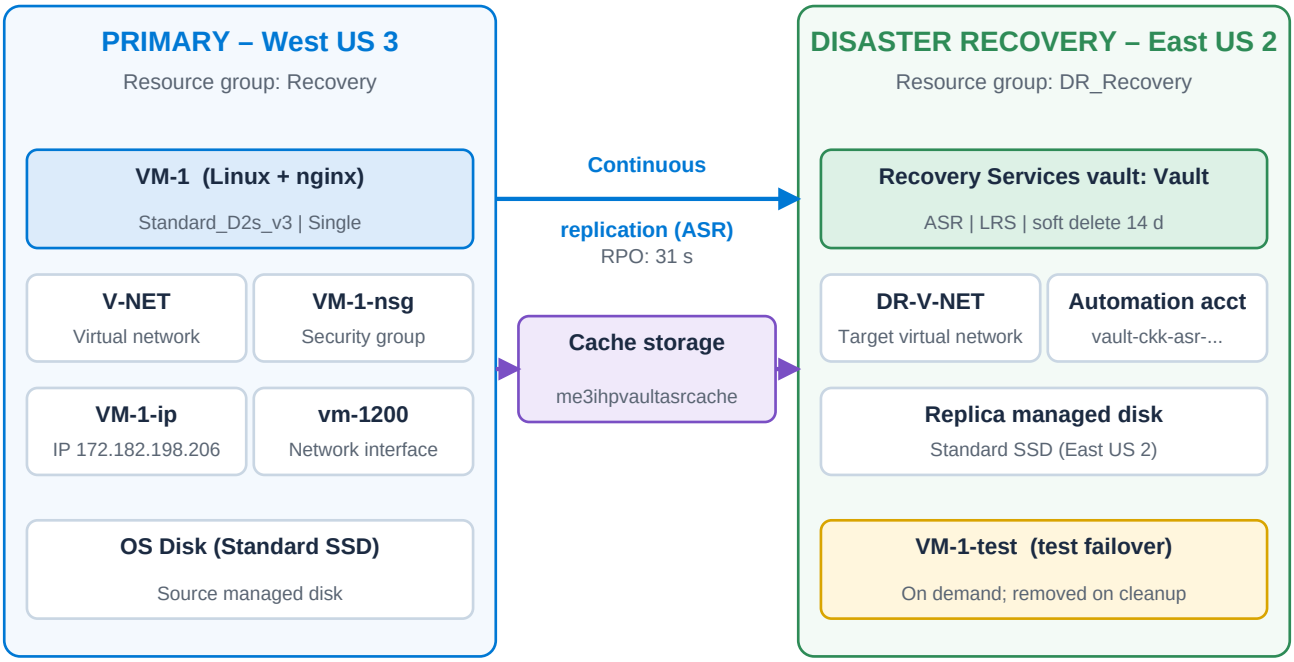


Figure 1 – Solution architecture (primary West US 3, DR East US 2).

3.1 How it works

- **Replication:** continuous, crash-consistent and periodic app-consistent recovery points are generated; the latest RPO is shown on the replicated item.
- **Test failover:** ASR builds a VM from a recovery point in the target region without stopping replication or touching the source VM.
- **Cleanup:** deletes the test VM and its resources, returning the item to its normal protected state.
- **Real failover:** promotes the replica to the live VM in East US 2; **re-protect** then reverses replication for failback.

4. Azure Services and Components

Component	Name / Value	Region	Purpose
Resource group (primary)	Recovery	West US 3	Contains the production workload
Virtual machine	VM-1 (Standard_D2s_v3)	West US 3	Linux web server running nginx
Virtual network	V-NET	West US 3	Primary network
Public IP	VM-1-ip (172.182.198.206)	West US 3	Internet access to the web page
NSG / NIC	VM-1-nsg / vm-1200	West US 3	Traffic filtering and network interface

Component	Name / Value	Region	Purpose
OS disk	VM-1_OsDisk_1_... (Standard SSD)	West US 3	Replicated source disk
Resource group (DR)	DR_Recovery	East US 2 (items)	Contains vault and DR resources
Recovery Services vault	Vault	East US 2	Orchestrates ASR; LRS; soft delete 14 days
Target virtual network	DR-V-NET	East US 2	Network for the recovered VM
Cache storage account	me3ihpvaultasrcache (Standard_LRS)	West US 3	Temporary store for replication data
Replica managed disk	VM-1_OsDisk_1_...asr (Standard SSD)	East US 2	Target copy of the OS disk
Automation account	vault-ckk-asr-automationaccount	East US 2	Agent update management for ASR
Test VM	VM-1-test (NIC vm-180-test)	East US 2	Created during test failover only

5. Implementation Steps

Step 1 – Primary environment and application

The primary workload was deployed in resource group **Recovery** in West US 3. It consists of the virtual network **V-NET**, the virtual machine **VM-1**, its public IP **VM-1-ip**, network security group **VM-1-nsg**, network interface **vm-1200** and the OS disk. nginx was installed on VM-1 and serves a simple page titled “Azure Disaster Recovery Project”, reachable over HTTP on the public IP.

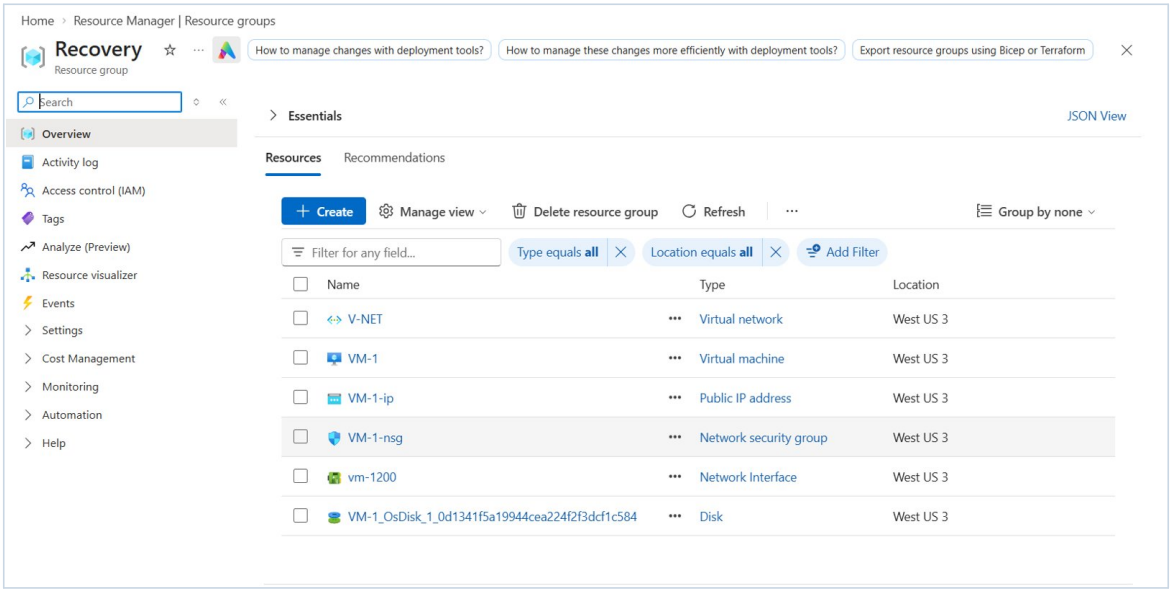


Figure 2 – Resource group “Recovery”: all primary resources in West US 3.

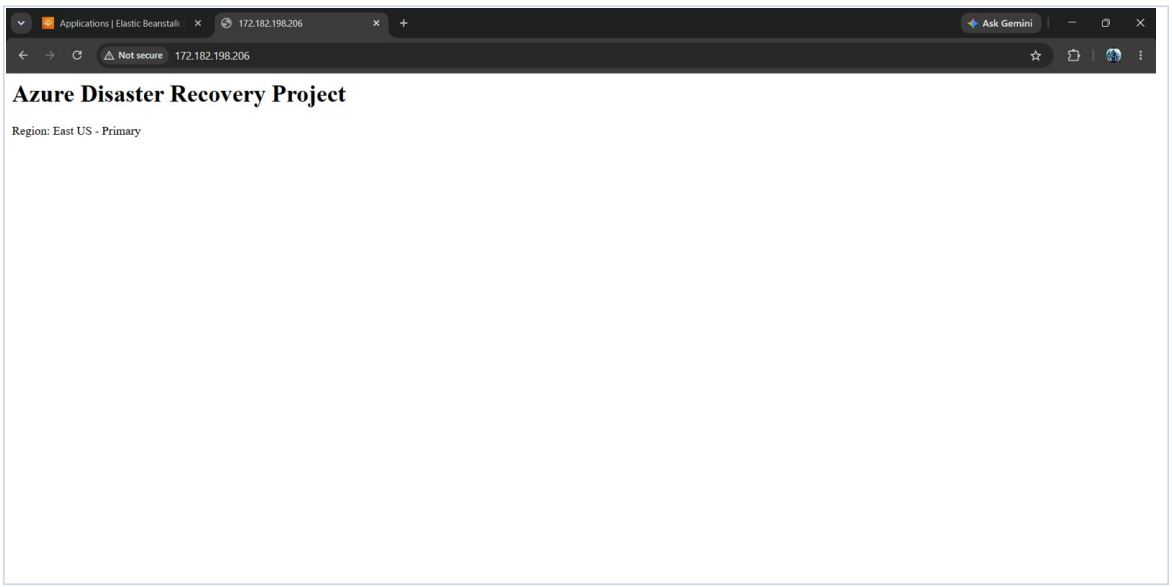


Figure 3 – Application page served by VM-1 from the primary environment.

Step 2 – Create the Recovery Services vault

A Recovery Services vault named **Vault** was created in the DR resource group **DR_Recovery** in **East US 2** (a different region from the source VM, as ASR requires). Settings: backup storage redundancy *Locally-redundant*, soft delete retention 14 days, immutability disabled, identities disabled, no customer-managed keys, and public network access enabled. An earlier vault deployment (*Recovery-vault*, started 12:04 AM) is also visible in the deployment history.

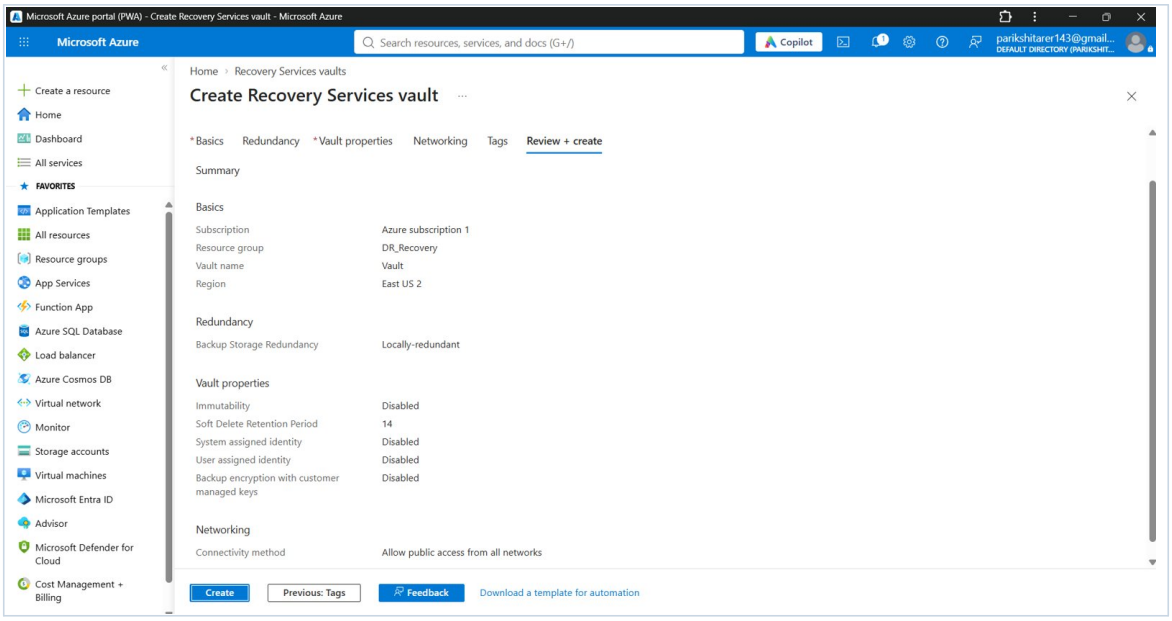


Figure 4 – Create Recovery Services vault: review + create summary.

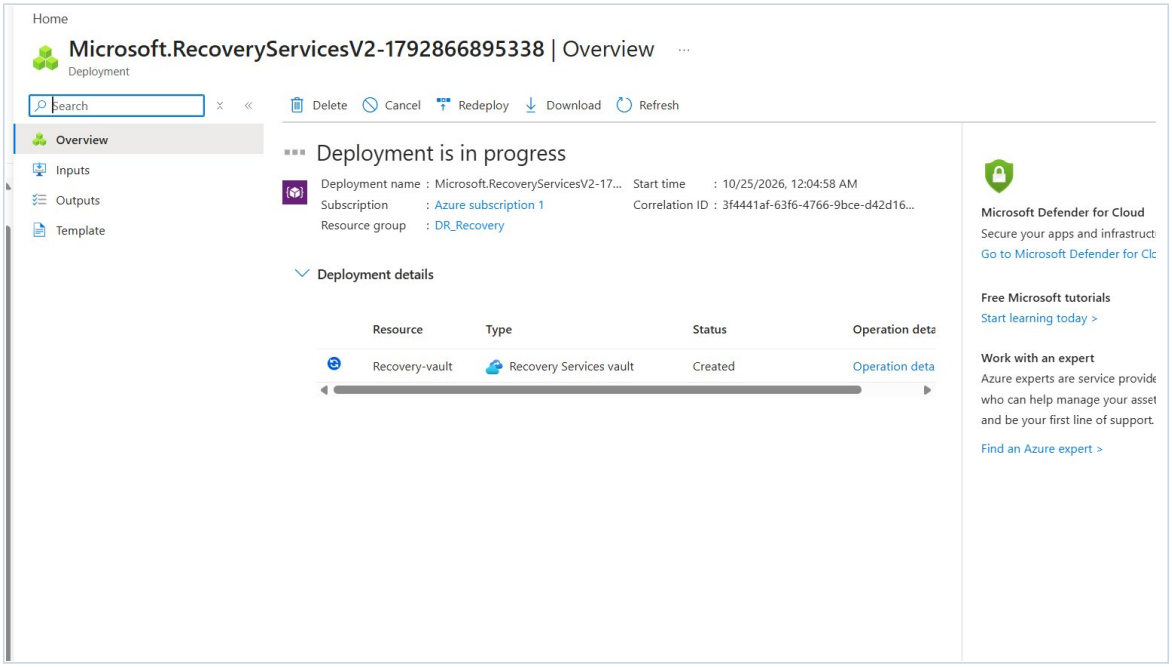


Figure 5 – Initial Recovery Services deployment (Recovery-vault) in DR_Recovery.

Step 3 – Configure disaster recovery for VM-1

From **VM-1** → **Backup + disaster recovery** → **Disaster recovery**, the target configuration was defined on the Advanced settings tab, then reviewed and started:

Setting	Source	Target
Subscription	Azure subscription 1	Azure subscription 1
VM resource group	Recovery	DR_Recovery
Virtual network	V-NET	DR-V-NET
Availability	Single instance	Single instance
Proximity placement group	Not applicable	None selected
Cache storage account	–	(new) me3ihpvaultasrcache [Standard_LRS]
Disk	Standard SSD (OS disk)	(new) replica managed disk, Standard SSD
Recovery Services vault	–	Vault (Azure subscription 1, DR_Recovery)

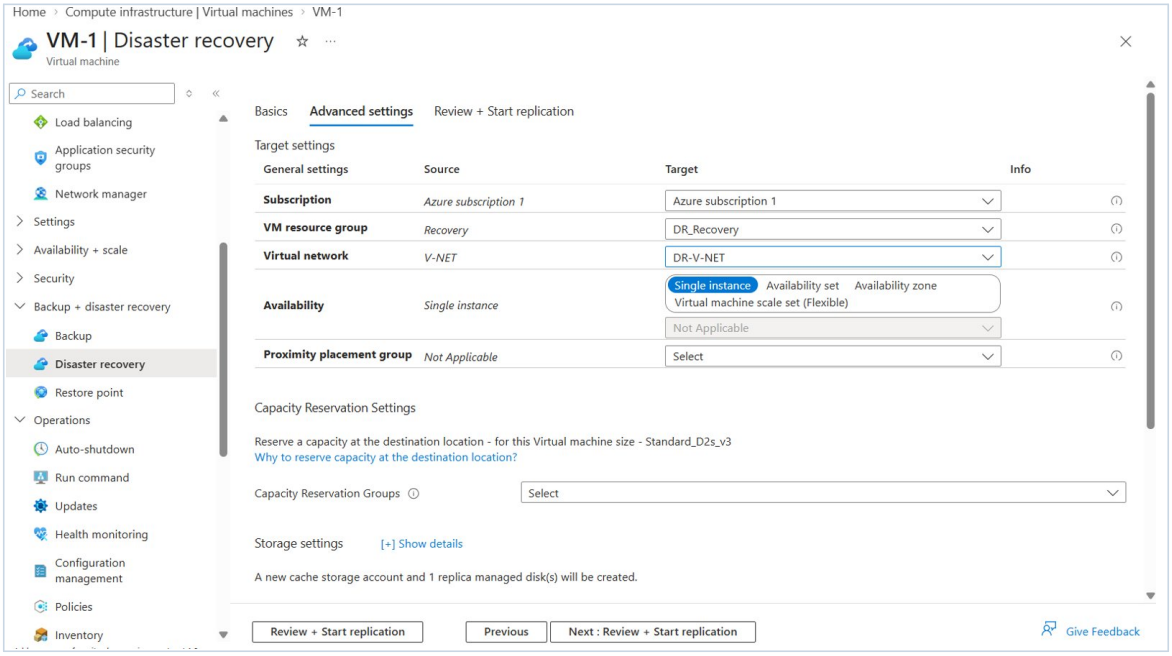


Figure 6 – Advanced settings: target subscription, resource group (DR_Recovery) and network (DR-V-NET).

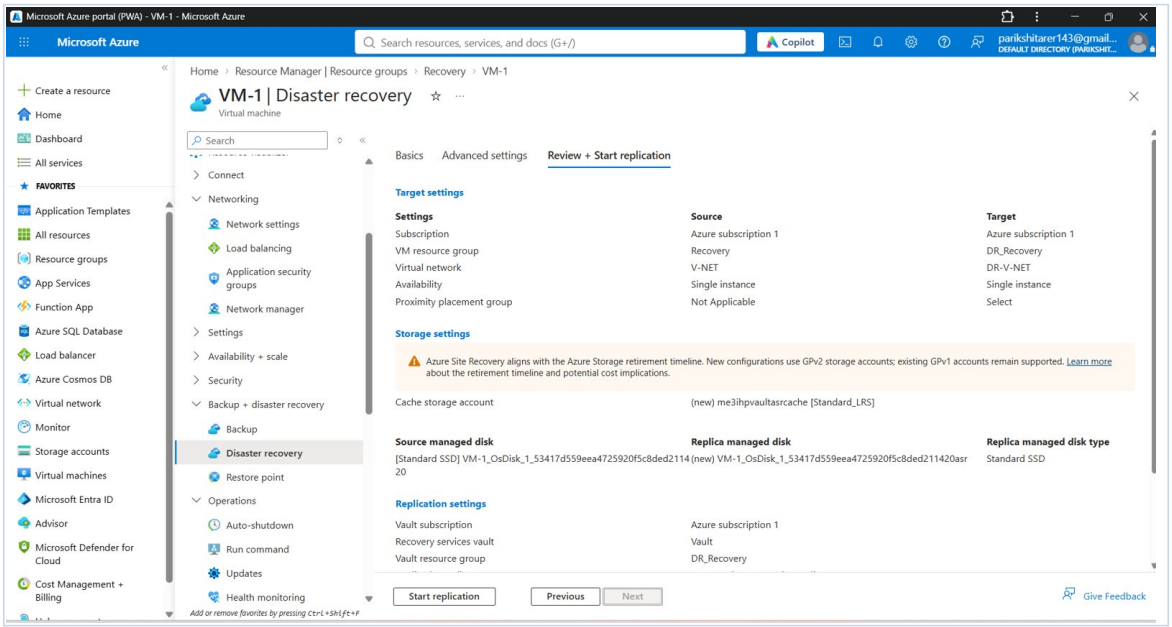


Figure 7 – Review + Start replication: target, storage and replication settings.

Starting replication triggered the **IaaSDR** deployment, which created the target-side resources (cache storage account, replica disk, automation account, ASR policies and the replication fabric). All listed deployment steps completed with status OK.

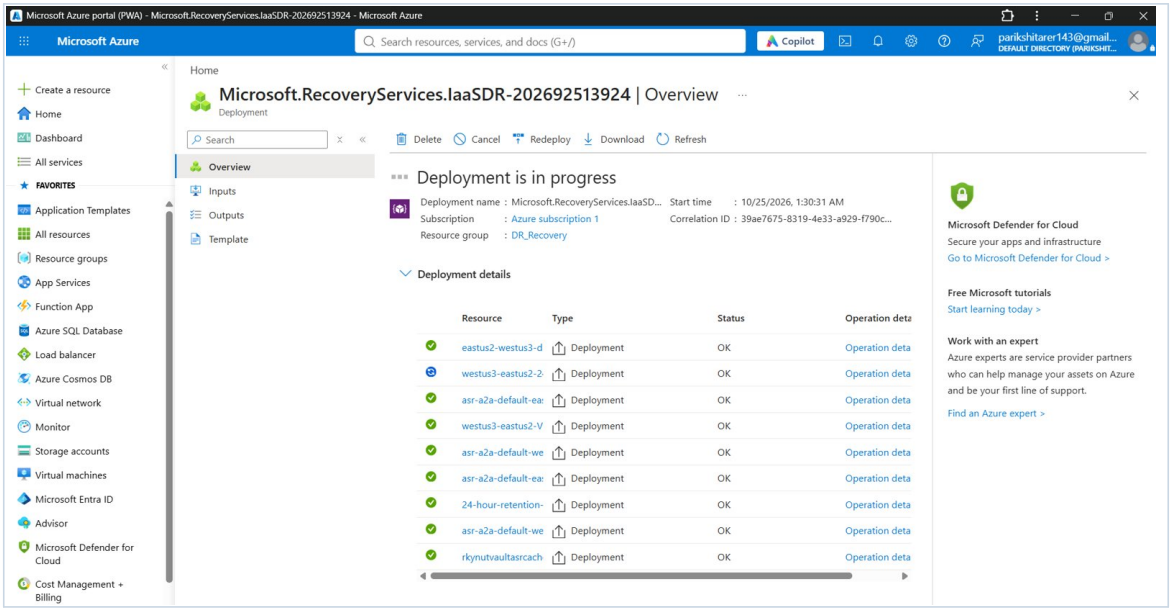


Figure 8 – IaaSDR deployment creating the ASR replication resources (started 1:30 AM).

Step 4 – Monitor initial replication

Immediately after enabling replication, the replicated item showed **Healthy** with **0% Synchronized** while the initial copy of the disk ran. The infrastructure view confirmed the chain VM → cache storage account → Azure Site Recovery → managed disk (1 disk).

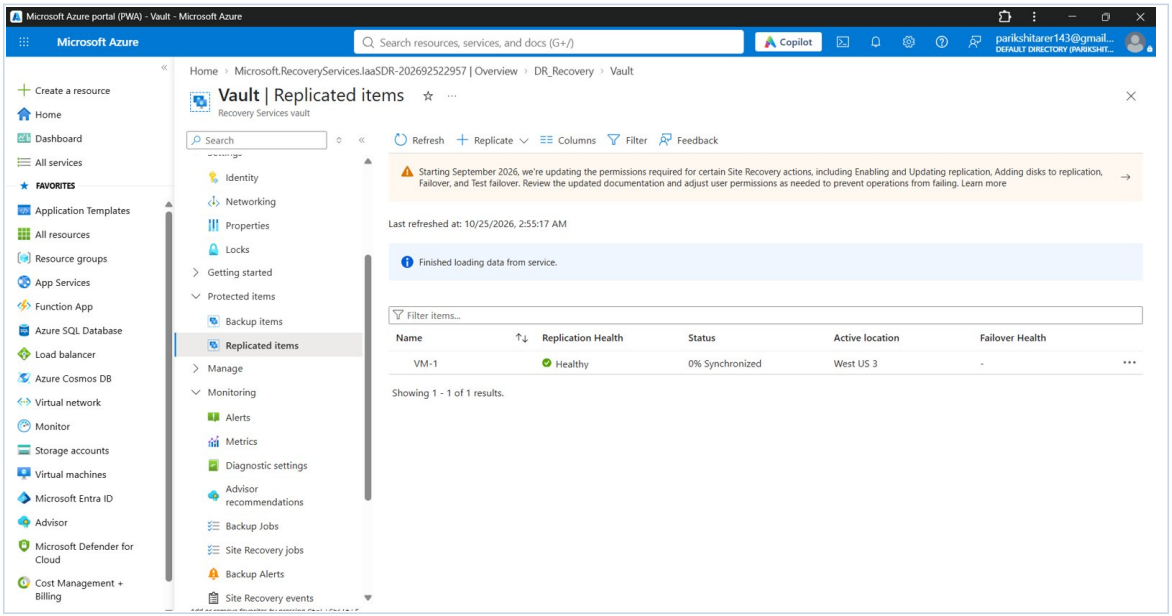


Figure 9 – Vault → Replicated items: VM-1 Healthy, 0% synchronized (2:55 AM).

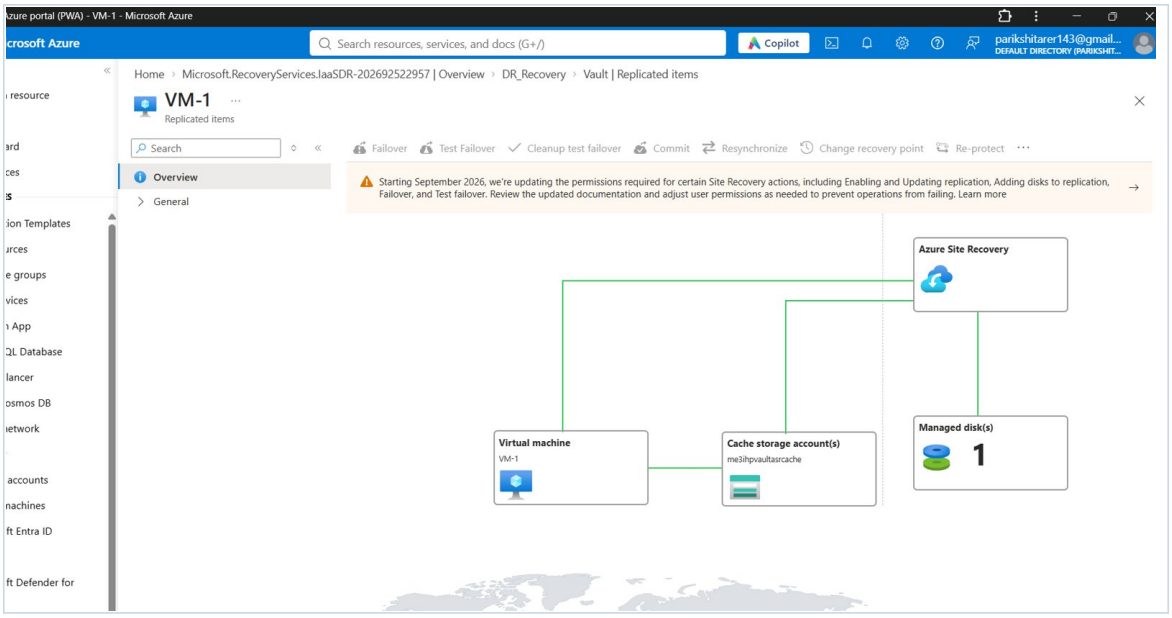


Figure 10 – Infrastructure view: VM-1, cache storage account me3ihpvaultasrcache, ASR and 1 managed disk.

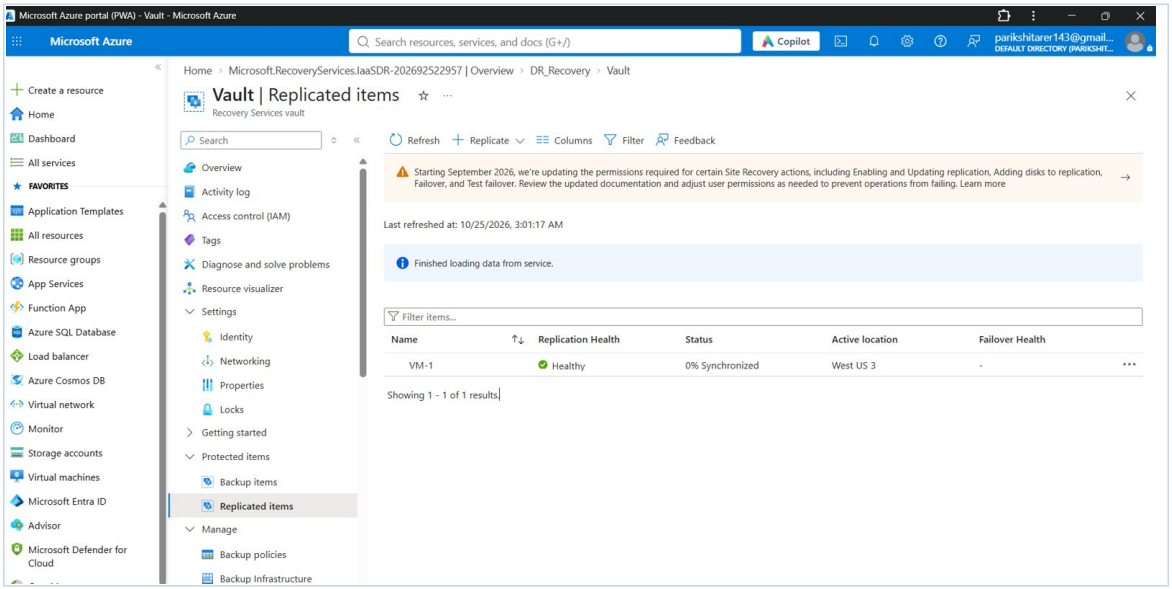


Figure 11 – Replicated items at 3:01 AM, still synchronizing.

The VM overview then showed *Waiting for first recovery point* with no configuration issues, agent version 9.67.7849.1 and agent status Healthy. A short time later the item moved to **Protected** with **RPO = 31 seconds** (as on 10/25/2026, 3:13:58 AM). Failover and Test Failover became available; the portal showed *Never performed successfully* for the last test failover, which is the expected state before the first drill.

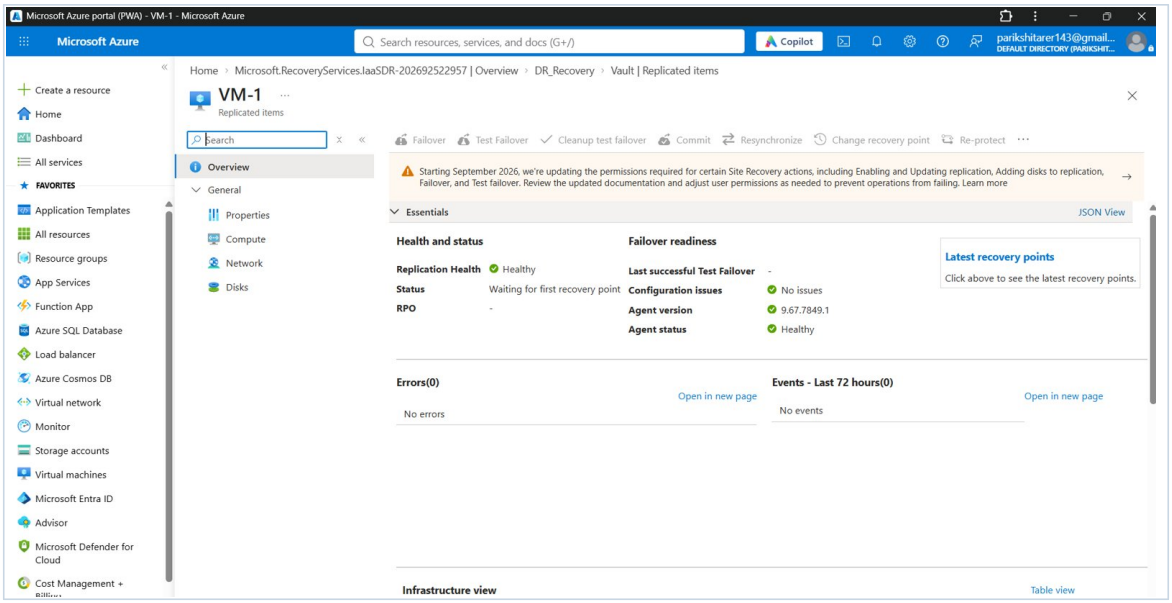


Figure 12 – Status: waiting for first recovery point; agent healthy, no errors.

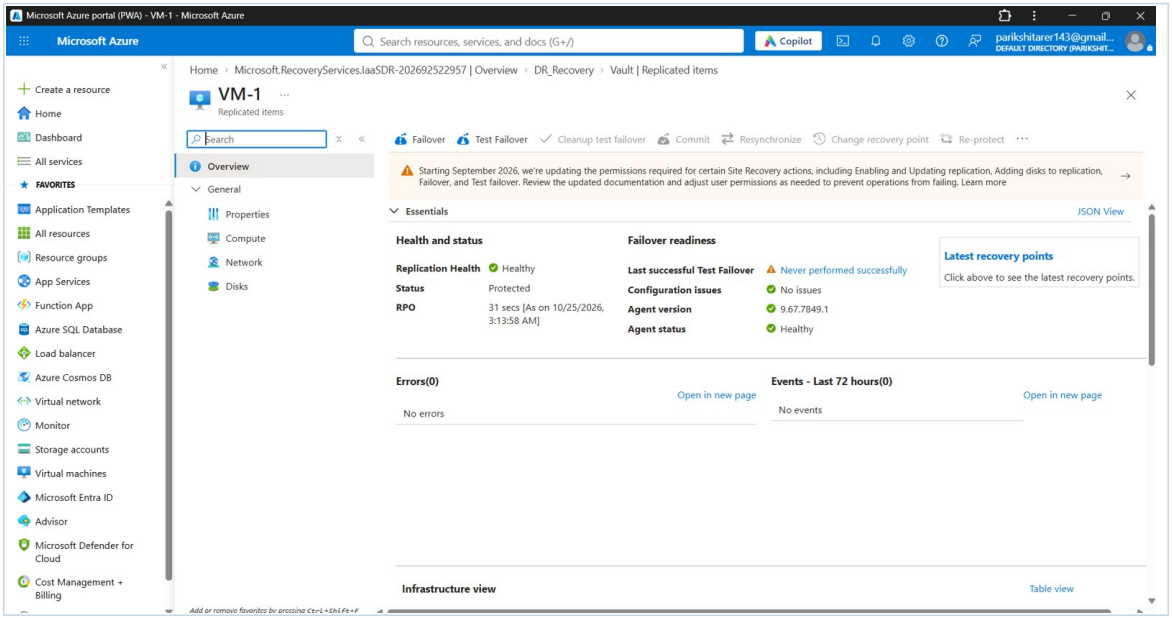


Figure 13 – Status: Protected, RPO 31 secs; Failover and Test Failover enabled.

Step 5 – Run a test failover

A test failover was started from the replicated item at 3:27:30 AM. ASR ran four jobs, all **Successful**:

Job step	Status	Start time	Duration
Prerequisites check for test failover	Successful	3:27:30 AM	00:00:00
Create test virtual machine	Successful	3:27:31 AM	00:01:37
Preparing the virtual machine	Successful	3:29:08 AM	00:00:36
Start the virtual machine	Successful	3:29:45 AM	00:00:01

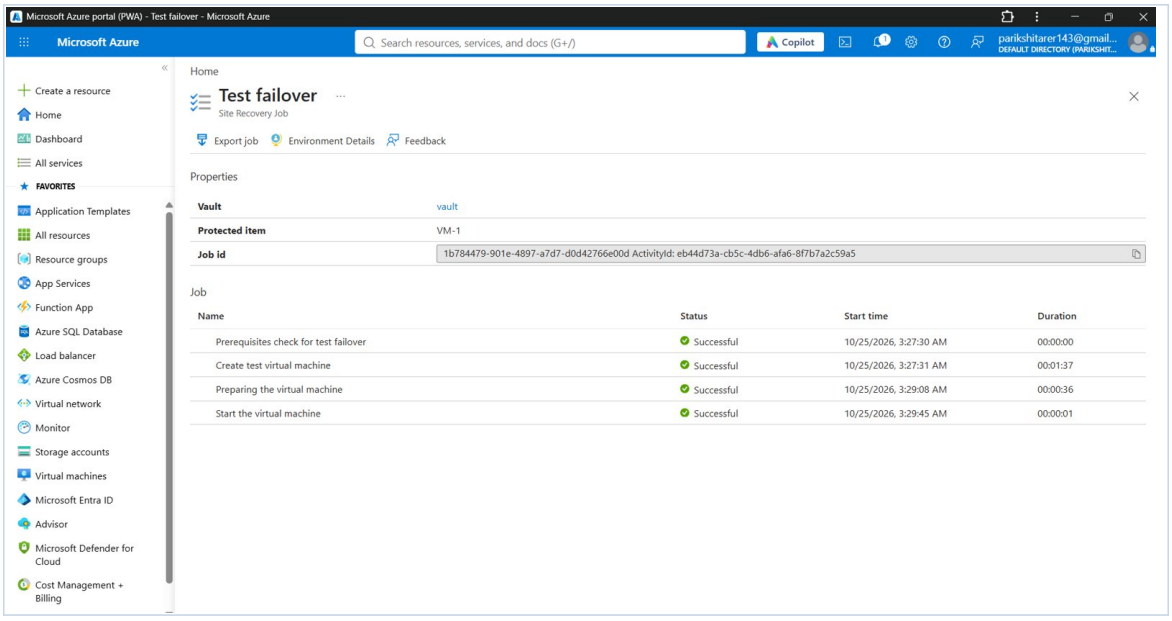


Figure 14 – Test failover job: all steps successful (total about 2 min 15 s).

The resource group **DR_Recovery** now contained the test VM **VM-1-test** (East US 2), its NIC **vm-180-test**, and the replica disks, next to the vault, automation account and cache storage account.

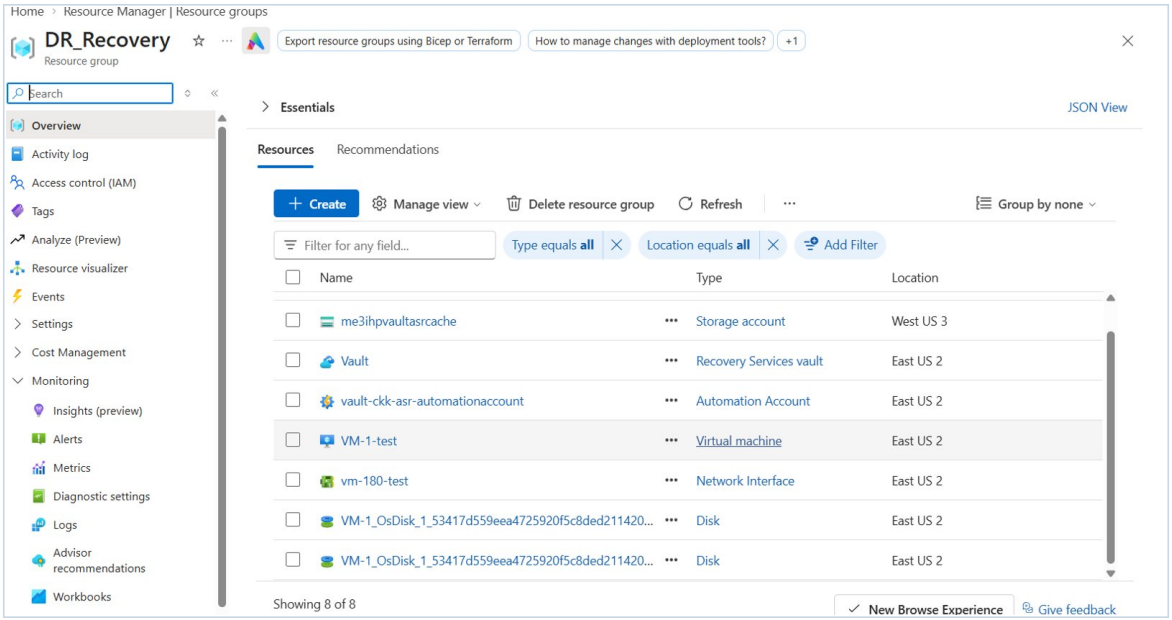


Figure 15 – DR_Recovery resource group during the test failover.

Step 6 – Validate the application on the DR VM

The test VM was opened through the **Serial console** and the web service was checked locally:

```
$ sudo systemctl is-active nginx
active
$ curl -I http://localhost
HTTP/1.1 200 OK
Server: nginx/1.20.1
Content-Type: text/html
Content-Length: 228
```

nginx started automatically from the replicated disk and returned the web page, confirming that the recovered VM is application-ready.

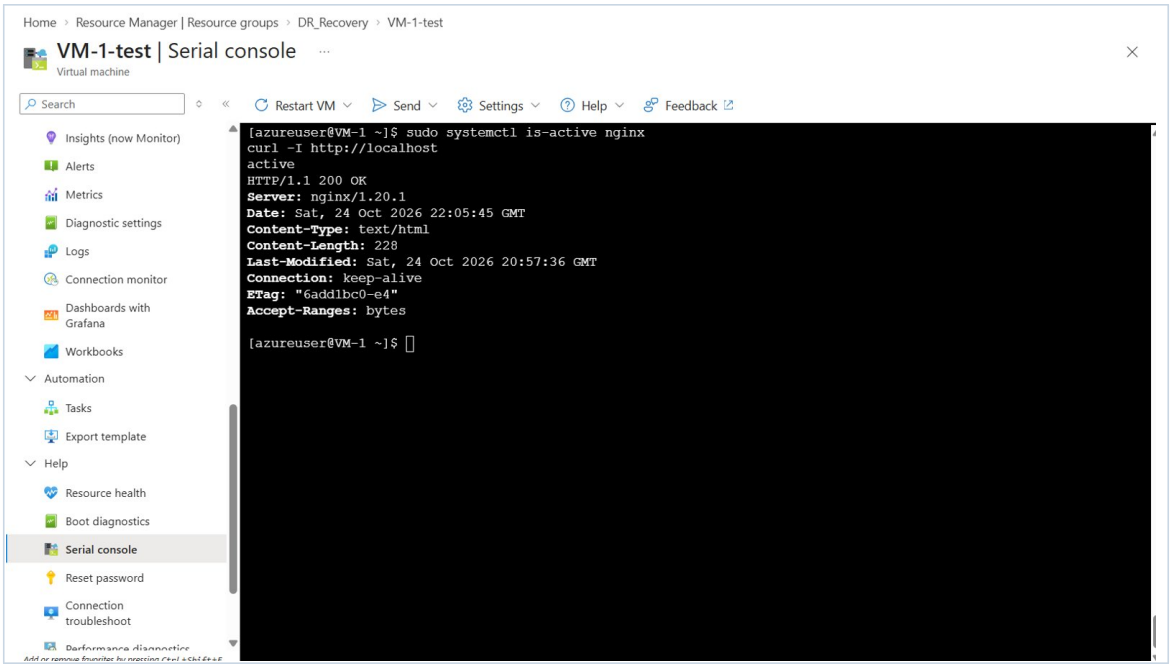


Figure 16 – Serial console on VM-1-test: nginx active, HTTP 200 OK.

Step 7 – Test failover cleanup

After validation, **Cleanup test failover** was run (3:42:04 AM). It checked prerequisites, ended the test failover and deleted the test virtual machine; all steps succeeded. This avoids ongoing compute cost and unblocks the next DR drill.

Job step	Status	Start time	Duration
Prerequisites check for test failover cleanup	Successful	3:42:04 AM	00:00:02
Ending test failover	Successful	3:42:07 AM	00:00:00
Clean up the test virtual machine	Successful	3:42:07 AM	00:01:09

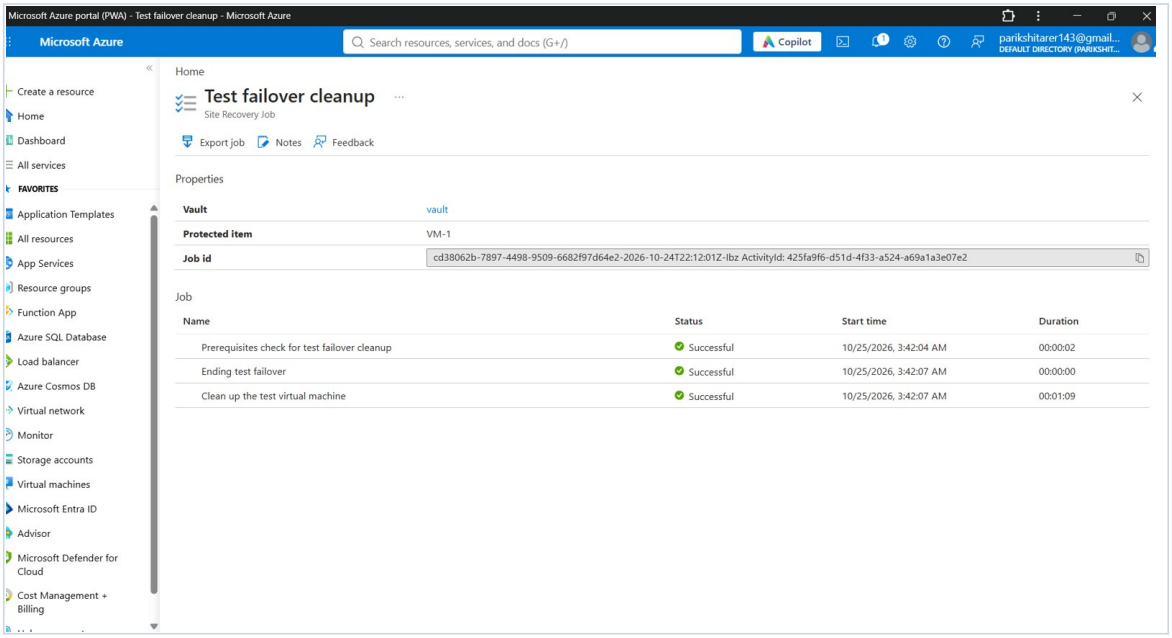


Figure 17 – Test failover cleanup job: all steps successful.

Step 8 – Post-test state

After cleanup the replicated item is again **Protected** with replication health **Healthy**, active location West US 3 and, for the first time, **Failover Health: Healthy** – evidence that the DR drill completed successfully.

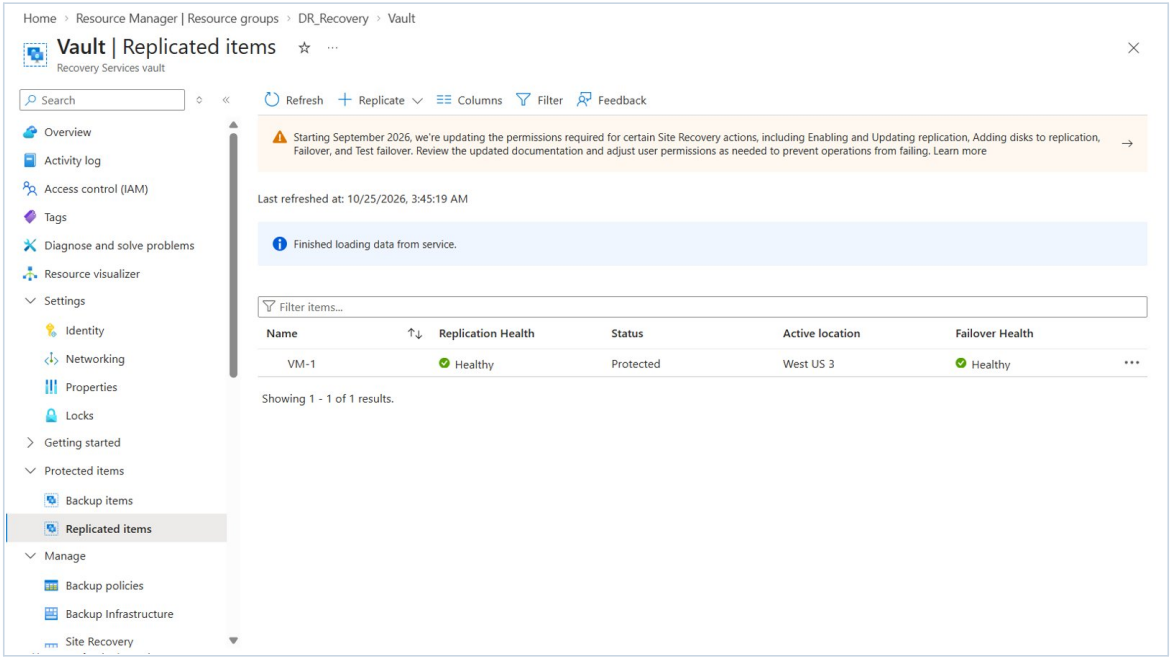


Figure 18 – Replicated items after cleanup: Protected, Healthy, Failover Health Healthy (3:45 AM).

6. Timeline and Results

All times are as displayed in the Azure portal on 25 October 2026.

Time	Event
12:04 AM	Recovery Services vault deployment started (Recovery-vault).
1:30 AM	Replication enabled; IaaSDR deployment created ASR resources.
2:55 AM – 3:01 AM	Initial synchronisation in progress (0% synchronized, Healthy).
~3:13 AM	Status Protected; RPO 31 s (as on 3:13:58 AM).
3:27:30 AM	Test failover started.
3:29:45 AM	Test VM VM-1-test started in East US 2.
~3:30 AM	nginx validated on test VM (active, HTTP 200 OK).
3:42:04 AM	Test failover cleanup started; completed in about 1 min 11 s.
3:45 AM	Final state: Protected, Healthy, Failover Health Healthy.

Key metrics

Metric	Observed value	Note
RPO	31 seconds	Maximum data-loss window at time of capture; varies with change rate
Initial replication	About 1 h 45 min	From enabling replication (1:30 AM) to Protected (~3:13 AM)
Test VM creation	About 2 min 15 s	Create + prepare + start jobs; indicates the platform part of RTO
Cleanup duration	About 1 min 11 s	Prerequisites + end test + delete test VM
Errors / events (72 h)	0 / 0	Shown on the replicated item overview

Note: a real-world RTO also includes decision time, DNS/traffic redirection and application checks, so it will be higher than the platform-only figure above.

7. Failover and Failback Runbook

Recommended operating procedure for a real regional outage (not executed in this project):

1. Declare the incident and confirm that the primary region (West US 3) is impaired.
2. Open **Vault** → **Replicated items** → **VM-1** → **Failover**. Select the recovery point (latest processed for fastest recovery, or latest app-consistent for data integrity).
3. Run the failover. Wait for the VM in East US 2 to start, then **Commit** the failover.
4. Attach a public IP / load balancer as needed, update DNS or Traffic Manager to the DR endpoint, and confirm the application responds.
5. When the primary region recovers, **Re-protect** the VM to replicate East US 2 → West US 3, then run a planned failover back and re-protect again to restore the original direction.

8. Observations and Recommendations

#	Observation	Recommendation
1	The application page reads “Region: East US – Primary”, but VM-1 runs in West US 3.	Correct the label to West US 3 (or read the region from instance metadata) to avoid confusion during a drill.
2	Validation of the DR VM was done locally via the serial console.	Also test from outside: assign a public IP and open port 80 in the NSG of the test network, or test from a jump VM in DR-V-NET.
3	The site is served over plain HTTP (browser shows “Not secure”).	Add HTTPS (certificate on nginx, or Application Gateway / Front Door).
4	DNS cut-over is not automated.	Use Azure Traffic Manager or Front Door and add the step to an ASR recovery plan.
5	Only one VM is protected, with no recovery plan.	Create a recovery plan (ordering, scripts) as more tiers are added.
6	Vault uses defaults: immutability off, public access, no identity or CMK.	Enable immutability, managed identity, private endpoints and RBAC for production use.
7	No alert rules are visible for RPO or replication health.	Configure Site Recovery alerts (RPO threshold, replication health, failover failures).
8	Portal banner: from September 2026 permissions for some Site Recovery actions change; storage retirement notice for GPv1 accounts.	Review the updated documentation and verify RBAC for operators; new setups already use GPv2 cache accounts.
9	An earlier vault deployment (Recovery-vault) appears in history.	Confirm no unused vault or resources remain, to avoid unnecessary cost.
10	DR drill performed once.	Schedule regular drills (for example quarterly) and record results.

9. Cost and Security Considerations

- **Ongoing costs while protected:** ASR per-instance fee, the replica managed disk, the cache storage account and replication transactions. The DR VM itself is **not** billed while it does not exist, which is why cleanup after each test matters.
- **Test failovers** incur compute and disk costs only until cleanup is run.
- **Security:** restrict NSG rules, keep the vault under least-privilege RBAC, keep soft delete enabled, and consider encryption with customer-managed keys for regulated data.
- Check current Azure pricing for the chosen regions and VM size before moving to production.

10. Challenges and Lessons Learned

- Initial synchronisation takes significant time (about 1 h 45 min here); plan for it before any demonstration or maintenance window.
- The first test failover shows a warning (“Never performed successfully”) until a drill completes – it is cleared by running one.
- Region labels on the application and in documentation must match the real deployment to keep drills unambiguous.
- The serial console is a quick, network-independent way to verify a recovered VM.

11. Conclusion

The project successfully delivers a working cross-region disaster recovery solution for an Azure IaaS workload. VM-1 is continuously replicated from West US 3 to East US 2 with an observed RPO of 31 seconds, a test failover produced a running, validated copy of the application in about two minutes, and cleanup returned the environment to a clean, protected state. With the recommended improvements (DNS automation, recovery plans, monitoring, vault hardening and periodic drills), the design can be taken from a validated proof of concept to a production-grade DR strategy.

Appendix A – Glossary

Term	Meaning
ASR	Azure Site Recovery – Azure service that replicates workloads to a secondary location and orchestrates failover.
RPO	Recovery Point Objective – maximum acceptable data loss, measured in time.
RTO	Recovery Time Objective – maximum acceptable time to restore service.
Recovery Services vault	Azure resource that stores replication configuration, recovery points and jobs.
Cache storage account	Storage account in the source region that temporarily holds replication data.
Test failover	Non-disruptive drill that creates a VM from a recovery point without affecting production or replication.
Commit / Re-protect	Commit finalises a failover; re-protect reverses replication direction to prepare failback.
Recovery plan	Ordered group of VMs and scripts failed over together.